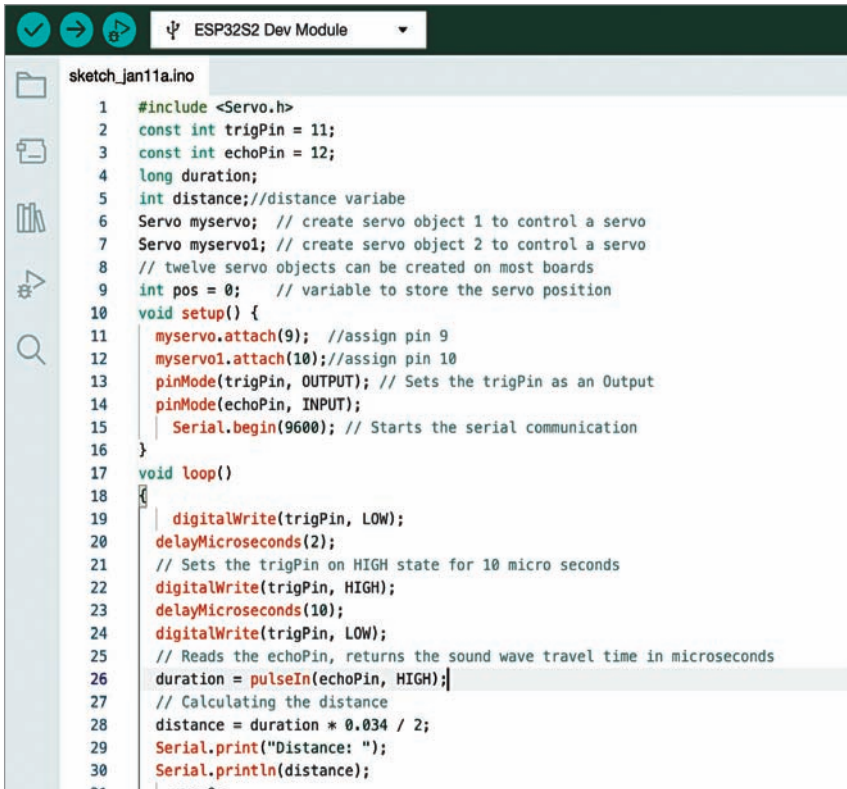




```

1  #include <Servo.h>
2  const int trigPin = 11;
3  const int echoPin = 12;
4  long duration;
5  int distance; //distance variable
6  Servo myservo; // create servo object 1 to control a servo
7  Servo myservo1; // create servo object 2 to control a servo
8  // twelve servo objects can be created on most boards
9  int pos = 0; // variable to store the servo position
10 void setup() {
11   myservo.attach(9); //assign pin 9
12   myservo1.attach(10); //assign pin 10
13   pinMode(trigPin, OUTPUT); // Sets the trigPin as an Output
14   pinMode(echoPin, INPUT);
15   Serial.begin(9600); // Starts the serial communication
16 }
17 void loop()
18 {
19   digitalWrite(trigPin, LOW);
20   delayMicroseconds(2);
21   // Sets the trigPin on HIGH state for 10 micro seconds
22   digitalWrite(trigPin, HIGH);
23   delayMicroseconds(10);
24   digitalWrite(trigPin, LOW);
25   // Reads the echoPin, returns the sound wave travel time in microseconds
26   duration = pulseIn(echoPin, HIGH);
27   // Calculating the distance
28   distance = duration * 0.034 / 2;
29   Serial.print("Distance: ");
30   Serial.println(distance);

```

Fig. 3: Snippet of the source code



Fig. 4: Mechanical assembly

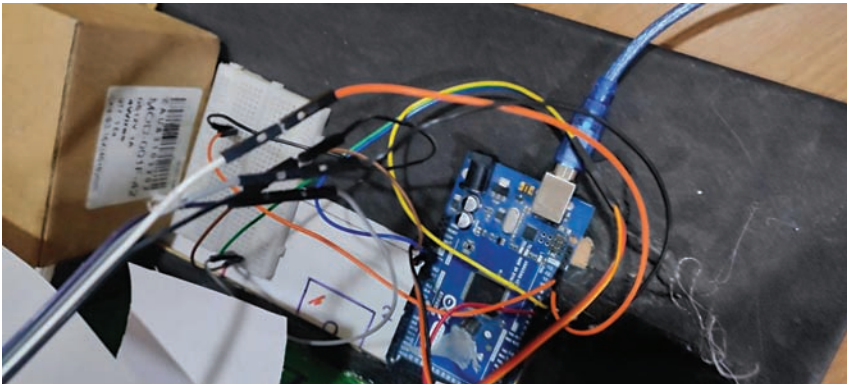


Fig. 5: Final assembly of the robot

EFY Note

The source code of this project is available for free download at www.electronicsforu.com

purposes, you can simply attach the paper-made hand to the servo shaft, as shown in Fig. 4.

Construction and testing

First, upload the source code into Arduino Uno using any desktop or laptop. Assemble the circuit on a breadboard or general-purpose PCB and complete the connections as per circuit diagram in Fig. 2. The mechanical assembly of both hands should be your choice, and the robot should be installed at the entrance gate so that when someone passes near the sensor, the robot arms and the hands get activated for greeting.

The working of the robot is simple. When anybody comes in front of the ultrasonic sensor within a distance of 100cm, the servo moves the hands to say 'Namaste.' When the person moves away from the robot, the robot's arms and hands move back to their normal position. The ultrasonic sensor acts as the eyes of the robot. Fig. 5 shows the final assembly of the robot, which has been tested in various situations and was found working efficiently. **EFY**



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